

Marco Aurelio Costa

Computer Vision & Machine Learning Engineer

Sausalito, CA 94965 | Marco.Costa.ML@Outlook.com | (415) 367-5652 | [LinkedIn](#) | [GitHub](#) | [HuggingFace](#)

EDUCATION

Chapman University | Orange, CA

M.S. Computer Science & Electrical Engineering

August 2024 - December 2025

Coursework: Statistical ML, Advanced AI, Bayesian Data Analysis, High Performance Computing, Database Systems

B.S. Computer Science, Minor: Information Security & Policy

August 2022 - May 2024

Coursework: Algorithm Analysis, Database Management, Numerical Analysis, Linear Algebra, Statistics

WORK EXPERIENCE

BioLumen - Bookkeeper & Operations Analyst | San Francisco, CA

May 2024 - Present

- Prepared balance sheets, profit-and-loss statements, and supporting financial records for external investors
- Monitored and reported Shopify sales, revenue, customer, and product analytics, highlighting performance trends.
- Developed financial projections and visualizations to support operational planning and comparison across business operations.
- Coordinated recurring financial disclosures with executives, legal counsel, accountants, and international investors.

RESEARCH EXPERIENCE

Faculty-Sponsored Independent ML Researcher | Chapman University

January 2026 - June 2026

- Conducted independent research with faculty sponsorship and access to Chapman's eight-GPU V100 computing cluster
- Designed an end-to-end research pipeline spanning computer vision, imitation learning, policy modeling and simulation
- Analyzed behavior cloning model performance and transitioned toward candidate ranking, simulation, PUCT, and MCTS
- Authored and submitted research manuscript describing pipeline to IEEE MLSP 2026 (under review, avg. reviewer rating 4/5)

COMPUTER VISION & VIDEO PROCESSING PROJECTS

Event Classification Tool for Sparse Video CV Pipelines

April 2026 - June 2026

TypeScript, React, Zod, C++, WebAssembly, Hyparquet

- Built a browser-based tool to define, debug, and label temporal events from sparse OCR and object-detection Parquet timelines
- Implemented manual classification and automated testing of classification configuration and metrics reporting
- Implemented WebAssembly + C++ acceleration for a subset of operations for efficiency boosts when classifying large datasets
- Automated export feature selection, reducing total parameters by ~80% for more efficient downstream DL model training

Raw Video to Structured State Reconstruction Pipeline

January 2026 - June 2026

Python, Linux, Bash, Jupyter, Ultralytics YOLOv8, PaddleOCR

- Built end-to-end imitation learning dataset on 500+ hours of downsampled video, generating 200M+ structured observations
- Trained object detection model achieving 0.995 mAP@0.5 on a synthetic dataset of 480K labeled instances across 12K images
- Trained character recognition model achieving 98.4% character accuracy on a synthetic dataset of 200K images
- Benchmarked OCR models and achieved 3.8x throughput improvement with <1% loss in character accuracy
- Ran distributed inference on machine learning cluster using 8 V100 GPUs to process 21M+ frames in under 60 hours

MACHINE LEARNING PROJECTS

AlphaZero-Style Balatro Agent Using PUCT + MCTS Simulator with Candidate Ranker

June 2026 - Present

C++, Python, XGBoost, PyTorch Transformers

- Trained XGBoost ranker on 24K expert demonstrations shrinking action space by 50-85% while retaining 95% candidate recall
- Rewrote the simulator and scoring logic in C++, cutting per-decision search time from minutes to seconds
- Prototype model defeated 1st and 2nd blind with 100% success rate using 2 of 23 action families suggesting sound architecture

SKILLS

- Programming: Python, C++, Java, SQL
- Machine Learning: PyTorch, Gymnasium, Imitation Learning, Gradient Boosting, Random Forest
- GPU / Parallel Computing: Multi-GPU Training, Distributed Inference, OpenMP, Open MPI
- Data & Systems: PySpark, Polars, Parquet, Multiprocessing, Data Pipelines
- Systems: AWS, Linux, Docker, Bash
- Computer Vision / OCR: OpenCV, YOLO, PaddleOCR, TrOCR, Object Tracking
- Languages: English (native), French (fluent), Italian (intermediate)